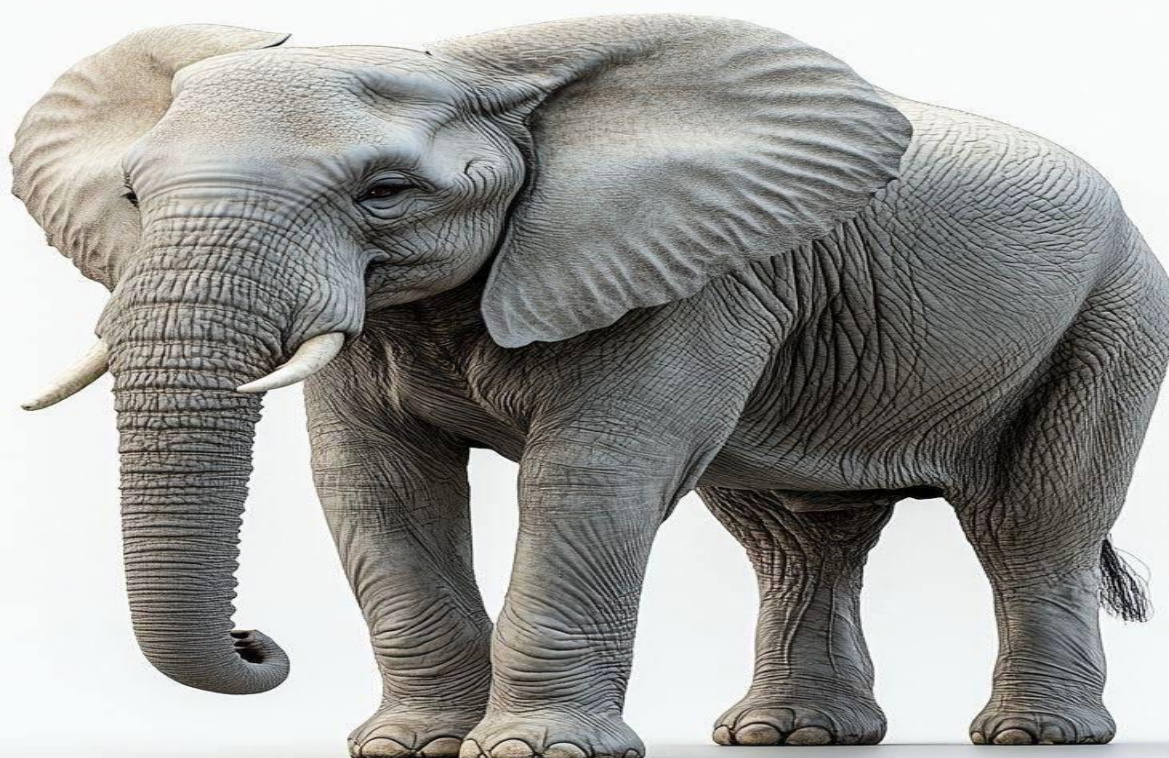




Elephant Herding Optimization (EHO)

Team Members:

- **Mohammed Emad Mohammed Elsafty**
- **Mohamed Magdy Abdel-Qader Mabrouk**
- **Giovanni Aaron Rawi Shafik**
- **Nesma Hesham Fathy**
- **Malak Ahmed Samih**
- **Menna Haitham Salah Zaki**

Under supervision:

Eng: Yousef Elbaroudy

Dr : Mahmoud Abdel-Aal



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Abstract about EHO:

Elephant Herding Optimization (EHO) is a population-based metaheuristic optimization algorithm that was originally proposed by Wang, Deb, and Coelho in 2015. The development of this algorithm was motivated by the increasing demand **for intelligent and robust optimization techniques** capable of solving complex real-world engineering and scientific problems. Although traditional optimization approaches are mathematically precise and well-established, they often encounter serious limitations when applied to problems characterized by high dimensionality, multiple local optima, discontinuities, or the absence of derivative information. In such challenging scenarios, nature-inspired optimization algorithms have demonstrated superior adaptability and flexibility, making them more effective in exploring large search spaces and obtaining near-optimal solutions within a reasonable computational time. EHO represents one of these **modern approaches**, offering a unique perspective inspired by the complex and highly organized social behavior of elephants.

The **biological inspiration of EHO** is derived from the natural social structure of elephant herds. In the wild, elephants live in well-organized family groups known as clans, where leadership is typically assumed by the matriarch, the oldest and most experienced female in the group. The matriarch plays a central role in guiding the clan, making critical decisions related to movement, safety, and access to essential resources such as food and water. Due to her accumulated experience and environmental knowledge, the matriarch is capable of leading the group efficiently across vast and challenging terrains. Other members of the clan closely follow her direction, continuously adjusting their positions in response to her movements. This coordinated and hierarchical behavior forms the foundation of the first mechanism in EHO, where candidate solutions are guided toward the best solution within a group.

Another important behavioral aspect observed in elephant societies is the natural separation of male elephants from their clans as they reach maturity. Adult males gradually leave their original family groups and either live independently or form smaller, less structured associations. This process plays a significant role in maintaining diversity within the elephant population and preventing inbreeding. In the context of optimization, this behavior is translated into a diversification strategy, where certain individuals are replaced with newly generated random solutions. This mechanism enhances the exploration capability of the algorithm and reduces the likelihood of premature convergence to suboptimal solutions, thereby improving overall search performance.

From a computational standpoint, the EHO algorithm begins by initializing a population of candidate solutions that are randomly distributed across the defined search space boundaries. The population is then divided into a number of clans, each containing a subset of individuals. Within each clan, the best-performing solution is identified as the matriarch. During each iteration, the positions of the elephants are updated according to **specific mathematical rules**. Non-matriarch individuals move toward both the matriarch and the center of their clan, influenced by a scaling parameter commonly denoted as α , which controls the degree of movement. Meanwhile, the matriarch updates her position by moving toward the clan center, guided by another parameter known as β . After completing the updating process, the worst-performing individual in each clan is replaced by a new randomly generated solution, simulating the natural separation process. This iterative procedure continues until a predefined stopping criterion, such as reaching a maximum number of generations, is satisfied. The best solution obtained throughout all iterations is then considered the final optimal or near-optimal result.

This report aims to provide a detailed and **well-structured presentation** of the Elephant Herding Optimization algorithm. It covers the biological inspiration behind the method, the mathematical formulation of its operators, and **the step-by-step implementation** procedure. In addition, a **numerical example is included** to demonstrate how the algorithm operates in practice, along with a comparative analysis against other widely used metaheuristic techniques. Through this comprehensive study, the report seeks to offer a clear and in-depth understanding of EHO and to highlight its significance as an effective optimization tool in modern computational applications.



Chapter (1): Introduction

Optimization problems appear in many real-world applications such as engineering design, machine learning, scheduling, routing, and resource allocation. These problems aim to find the best possible solution among a large set of possible alternatives while satisfying certain constraints. Traditional optimization techniques often struggle when dealing with complex, nonlinear, or high-dimensional problems. As a result, researchers have developed *metaheuristic algorithms* inspired by natural phenomena to solve these difficult optimization challenges more efficiently.

1.1 The Elephant Herding Optimization (EHO) is a swarm intelligence-based *metaheuristic* optimization technique inspired by the social behaviour of *elephant herds in nature*. In the wild, elephants live in groups called *clans*, which are led by a matriarch. The members of each clan move together, share information, and adjust their positions according to the leader. Additionally, male elephants often leave their clans and live independently, which increases diversity within the population. These natural behaviors have been mathematically modeled to create an optimization algorithm that balances *exploration* and *exploitation* in the search space.

1.2 The main objective of the Elephant Herding Optimization algorithm is to find **the optimal or near-optimal solution for complex optimization problems**. It achieves this by *simulating the movement of elephants within clans* and **updating** their positions based on **the best solutions found so far**. The algorithm uses two main operators: **clan updating and separating**. The clan updating operator improves solutions by moving individuals toward the clan leader, which **enhances exploitation**. On the other hand, the separating operator replaces some individuals with new random solutions, which **increases exploration** and prevents premature convergence. This combination helps the algorithm avoid **local optimal** and **improve global search performance**.

1.3 The problem domain of the Elephant Herding Optimization algorithm is very broad. It can be applied to continuous optimization problems, discrete optimization problems, and combinatorial optimization tasks. In engineering, it is used for parameter tuning, structural design optimization, and control system optimization. In machine learning, it is applied for **feature selection, neural network training, and hyperparameter optimization**. In operations research, it can solve scheduling problems, routing problems, and resource allocation tasks. The flexibility of the algorithm makes it *suitable for solving both single-objective and multi-objective optimization problems*.

Another important aspect of the Elephant Herding Optimization algorithm is its **1.4 scalability in high-dimensional search spaces**. Many optimization techniques face performance degradation as the number of dimensions increases. However, **EHO** maintains a balance between exploration and exploitation, allowing it to handle large-scale problems effectively. The dimensions in the optimization problem represent the variables that define a candidate solution. Each elephant in the population represents a potential solution, and each dimension corresponds to *a parameter being optimized*. The algorithm iteratively updates these dimensions to improve solution quality.

The Elephant Herding Optimization algorithm also has several **advantages**. **It is simple to implement, requires few control parameters, and demonstrates strong global search capability**. Additionally, it provides good convergence speed and *avoids stagnation in local minima*. However, like other metaheuristic algorithms, its performance depends **on parameter tuning and problem characteristics**. Therefore, hybrid approaches combining EHO with other optimization techniques have been proposed to enhance its **efficiency**.

Despite the strong performance and wide applicability of Elephant Herding Optimization, the algorithm is not without its weaknesses. Like all metaheuristic methods, EHO carries a set of inherent limitations that researchers and practitioners must be aware of when deciding whether it is the appropriate tool for a given problem.



Parameter Sensitivity One of the most frequently noted challenges associated with EHO is its sensitivity to the choice of control parameters. The scaling factors alpha and beta, the number of clans, and the population size all have a direct and significant influence on the quality of the solutions produced and the speed at which the algorithm converges. There is no universal set of values that works well across all problem types, which means that users are often required to invest considerable time and computational effort in tuning these parameters before the algorithm can deliver reliable results. This trial-and-error process adds overhead and reduces the plug-and-play convenience that practitioners often look for in optimization tools.

Slow Convergence on High-Dimensional Problems While EHO performs well on low to medium dimensional problems, its convergence tends to slow down noticeably as the number of dimensions increases. In very high-dimensional search spaces, the clan-based update mechanism may struggle to explore the space efficiently, leading to longer runtimes and a higher risk of settling for suboptimal solutions. This scalability issue limits the direct applicability of the standard EHO algorithm to problems that involve a very large number of decision variables without additional modifications or hybridization strategies.

Risk of Premature Convergence Although the separating operator is specifically designed to maintain population diversity, it does not always provide sufficient protection against premature convergence, especially when the population size is small or the number of clans is large relative to the total number of individuals. In such configurations, the clans may lose internal diversity too quickly, causing the algorithm to converge prematurely around a local optimum rather than continuing to explore the broader search space in search of the global optimum.

In conclusion, the Elephant Herding Optimization algorithm is a powerful swarm intelligence-based optimization technique inspired by **elephant social behavior**. Its objective is to efficiently search for optimal solutions in *complex and high-dimensional problem spaces*. Due to its flexibility and effectiveness, it has been widely applied in **engineering, machine learning, and operational optimization domains**. The algorithm's ability to balance exploration and exploitation makes it a valuable tool for solving **challenging optimization problems**.

Classification of Elephant Herding Optimization (EHO)

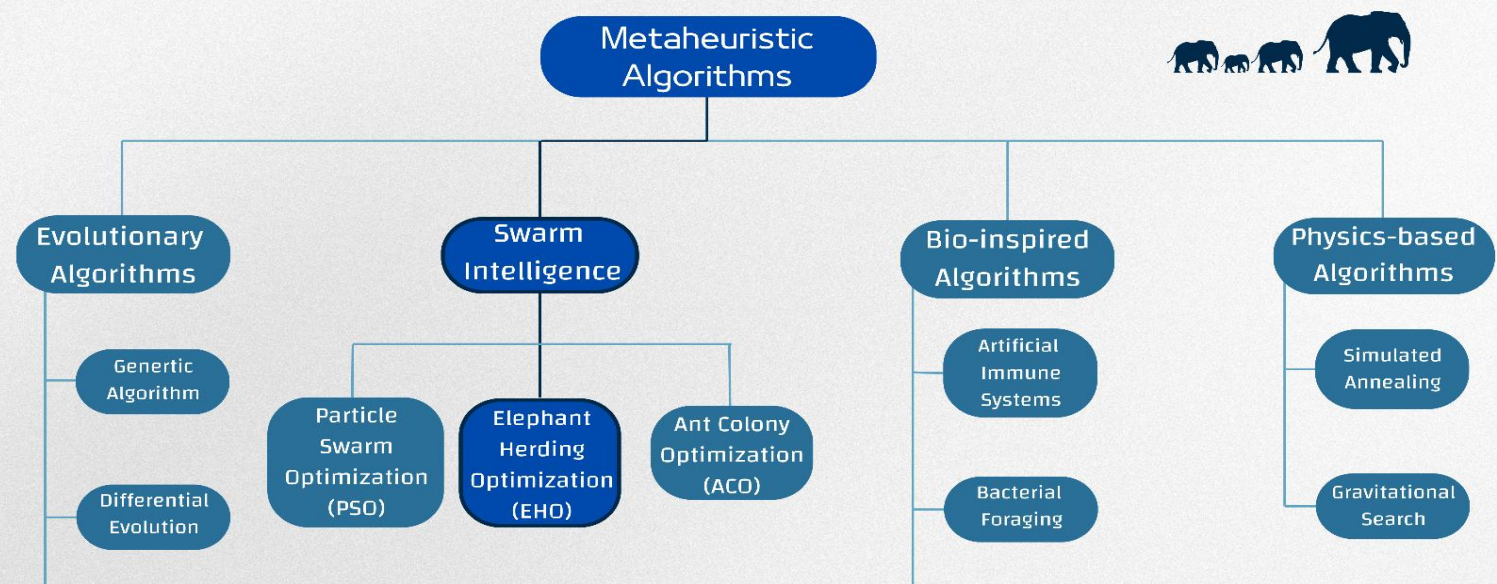


Figure1-EHO Classification



Chapter (2): Literature Review

Elephant Herding Optimization has been adopted by researchers across a broad spectrum of scientific and engineering disciplines since its introduction. Its ability to balance global exploration with local exploitation, combined with its straightforward implementation, made it an attractive choice for tackling complex real-world optimization challenges. The studies presented in this chapter reflect the diversity of domains in which EHO has been successfully applied, ranging from healthcare and power systems to finance and wireless communications, each demonstrating the practical value and adaptability of the algorithm beyond theoretical benchmarks.

2.1 Application1: Cancer Prediction Using EHO-Based Neural Network

One of the most impactful applications of EHO in healthcare is cancer prediction and classification. Researchers developed a nature-inspired machine learning model using the EHO algorithm to optimize a neural network for classifying different cancer types, including lung cancer, breast cancer, and cervical cancer. In this study, feature selection was first performed using ANOVA and Kruskal-Wallis statistical tests to identify the most relevant features from high-dimensional cancer datasets. The EHO algorithm was then used to train and optimize the neural network weights, guiding the search for the best parameter configuration across the solution space.

The algorithm's clan-based exploration allowed it to efficiently scan the feature space while avoiding local optima — a common challenge in high-dimensional biomedical datasets. Performance was evaluated using Root Mean Square Error (RMSE) and Correct Classification Rate (CCR), and results were compared against the Local Linear Wavelet Neural Network (LLWNN) and Particle Swarm Optimization (PSO). The EHO-based model demonstrated superior classification accuracy, highlighting the algorithm's strong ability to optimize complex neural architectures for medical diagnosis tasks.

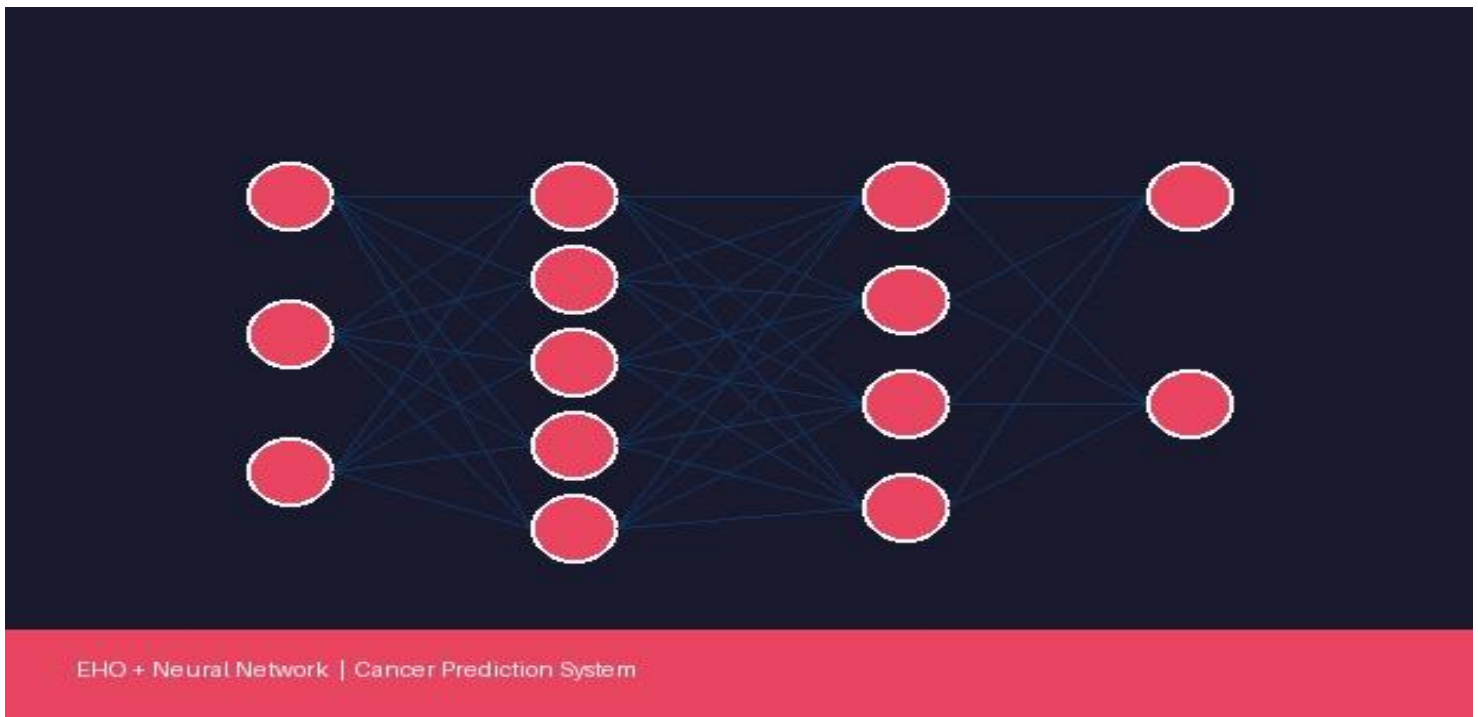


Figure 2.1 – EHO-optimized neural network architecture applied to multi-type cancer classification and prediction.



2.2 Application 2: Smart Home Energy Management System (HEMS)

With the rise of smart grids and time-of-use (TOU) electricity pricing, researchers have applied EHO to address the appliance scheduling problem in Home Energy Management Systems (HEMS). In a notable study by Sarwar et al. (2018), EHO was used to schedule 12 appliances in a simulated smart home, categorized as either interruptible or non-interruptible loads. The goal was to minimize electricity cost while reducing user waiting time.

The EHO algorithm modeled each possible appliance schedule as a 'position' in the search space, with clans of candidate solutions evolving towards cost-optimal schedules under the guidance of a matriarch solution. Simulation results confirmed that EHO outperformed the Enhanced Differential Evolution (EDE) algorithm and the unscheduled baseline, achieving substantial cost reduction without significantly increasing user discomfort. A similar study by Chibani & Ouhadda (2022) further validated EHO's effectiveness in smart city energy scenarios, showing meaningful reductions in electricity bills through intelligent load shifting.



Figure 2.2 – EHO-based appliance scheduling framework for smart home energy cost minimization using TOU pricing.

2.3 Application 3: Wireless Sensor Network (WSN) Node Localization

Accurate node localization is a fundamental challenge in Wireless Sensor Networks (WSN), where GPS cannot be deployed on every sensor. Strumberger et al. (2018) proposed the use of EHO to solve the WSN localization problem by treating unknown node positions as optimization variables. The algorithm minimized the localization error by iteratively updating estimated positions of unknown nodes based on known anchor positions and received signal strength. EHO's two operators played distinct roles: the clan updating operator performed coordinated exploitation around the best-known position estimates (the matriarch), while the separating operator ensured sufficient diversity to escape local minima. Experimental results showed EHO achieving competitive localization accuracy compared to other swarm-based methods including the Monarch Butterfly Optimization algorithm. The study confirmed EHO's suitability for real-time network deployment scenarios where computational efficiency and accuracy are both critical.

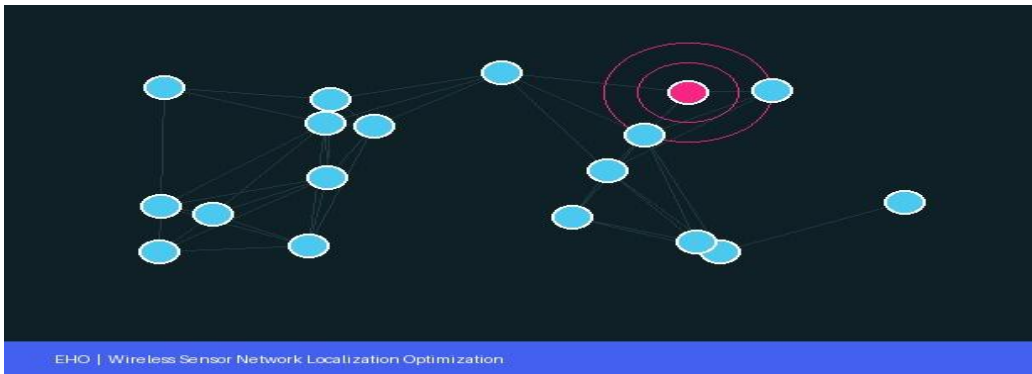


Figure 2.3 – EHO-optimized sensor node localization in a distributed Wireless Sensor Network topology.



2.4 Application 4: Optimal Virtual Power Plant (VPP) Scheduling

Virtual Power Plants (VPPs) aggregate distributed energy resources — including renewable sources, battery storage, and controllable loads — into a unified management system. Cahig et al. proposed a decision support tool based on EHO to solve the day-ahead scheduling problem for VPPs, with the objective of minimizing total 24-hour operating costs including energy losses.

Each 'elephant' in the EHO search represented a candidate energy dispatch schedule across the planning horizon. The matriarch guided the clan toward cost-efficient solutions, while the separating operator introduced variability to explore alternative dispatch strategies. Results showed that the canonical EHO variant yielded optimal or near-optimal scheduling outcomes, outperforming baseline methods. A related application by Parashar et al. further extended EHO to handle uncertain renewable generation in microgrids, confirming the algorithm's robustness under stochastic energy conditions. EHO also demonstrated strong performance in optimizing PI controller gains for STATCOM-based reactive power compensation in hybrid PV-wind systems.



Figure 2.4 – EHO applied to Virtual Power Plant scheduling integrating solar, wind, and storage energy resources.

2.5 Application 5: Travelling Salesman Problem (TSP) Route Optimization

The Travelling Salesman Problem (TSP) is a classic NP-hard combinatorial optimization problem requiring the identification of the shortest route visiting a set of cities exactly once. Hossam et al. (2019) and Almufti et al. (2019) applied discrete variants of EHO to solve both symmetric and general TSP instances, adapting the continuous position-update operators to work within permutation-based solution representations.

In the discrete EHO formulation, each elephant's position encoded a city visitation sequence, and the matriarch-guided clan updating operator performed permutation crossover operations to generate improved routes. The separating operator introduced random route permutations to maintain population diversity and prevent premature convergence. Comparative studies against U-Turning Ant Colony Optimization (U-TACO) and Discrete Cuttlefish Optimization showed EHO achieving competitive solutions on standard TSP benchmarks, validating the algorithm's adaptability to combinatorial domains beyond continuous optimization.



Figure 2.5 – EHO-generated optimal tour for the Travelling Salesman Problem across a set of 12 benchmark cities.



2.6 Application 6: Large-Scale Social Media Information Access

A highly novel application of EHO was proposed by Drias et al. (2024), who developed an enhanced EHO framework for information retrieval on large-scale social media platforms, specifically targeting over 1.4 million tweets. Inspired by Information Foraging Theory (IFT), the approach modeled the process of finding relevant information as an optimization problem, where each elephant represented a candidate information-gathering strategy navigating a vast data landscape.

To handle the scale of social media data, the authors introduced new operators into the standard EHO framework: territory delimitation and clan migration using clustering techniques. These additions enabled the algorithm to partition the search space into manageable regions and relocate entire clans to more information-rich territories, mirroring how elephants migrate seasonally. Experiments demonstrated that the enhanced EHO significantly outperformed the original algorithm and achieved superior results compared to Ant Colony System and Particle Swarm Optimization on the same information access tasks, confirming EHO's potential for large-scale data-driven applications.

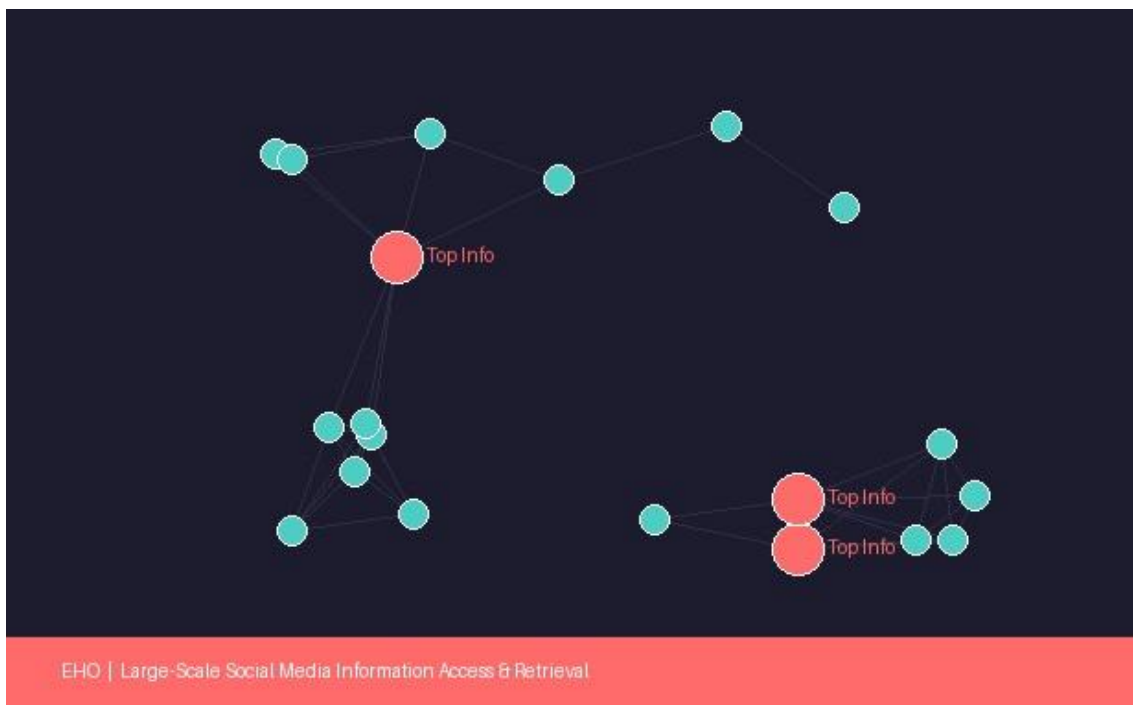


Figure 2.6 – Enhanced EHO with territory delimitation applied to large-scale social media information retrieval and ranking.

Summary

The reviewed studies collectively demonstrate that the Elephant Herding Optimization (EHO) algorithm is a highly versatile and competitive metaheuristic. Its applications span medical diagnosis, energy management, network engineering, combinatorial optimization, and large-scale data retrieval. The algorithm's dual-operator architecture — combining coordinated clan-based exploitation with diversity-preserving separation — makes it well-suited for a broad range of problem domains. The following chapter builds upon these findings to present the proposed application of EHO in the current research project.





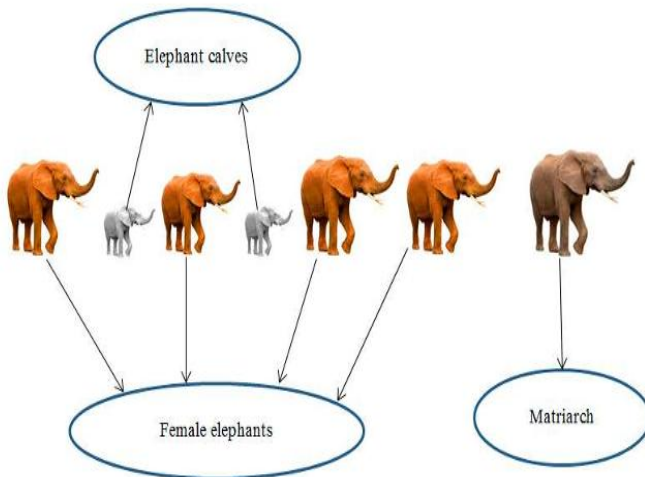
Chapter (3): The Proposed Algorithm

3.1 Natural Behavior of Elephants (Inspiration Source) :

Elephants live in a highly organized matriarchal society, which forms the core inspiration for the algorithm.

In nature, elephants are structured into small family groups called **clans**. Each clan is led by the oldest and most experienced female known as the **matriarch**. The matriarch guides the entire clan toward food sources, water, and safe routes using her vast knowledge and memory. All other members of the clan (mainly adult females and calves) follow the matriarch closely, moving toward her position to ensure group protection and efficient foraging. This collective movement toward the leader promotes strong group cohesion and local exploitation of the best-known resources.

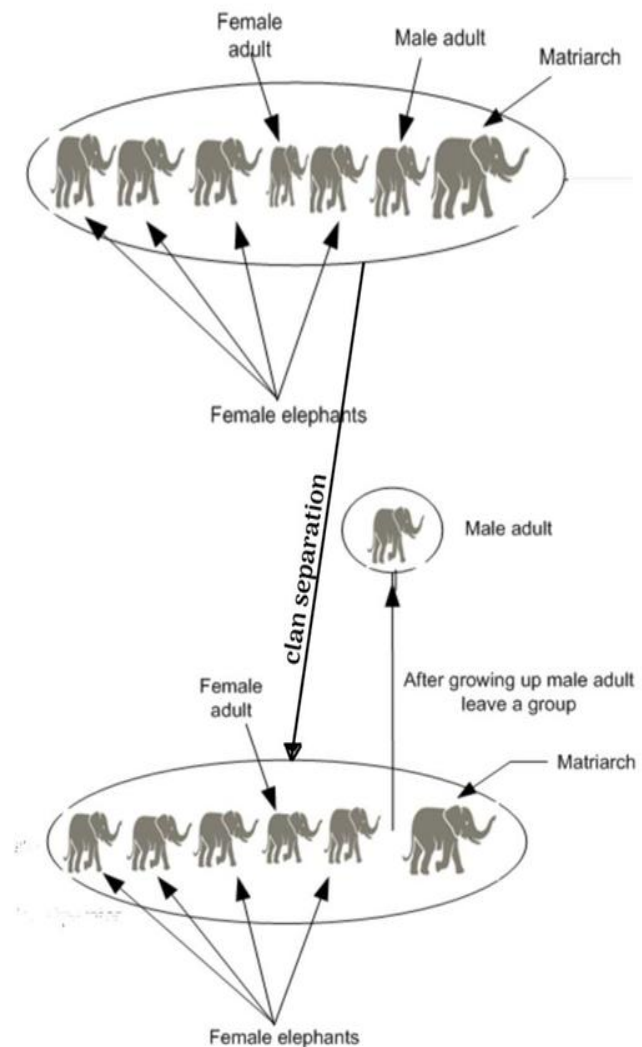
Additionally, the matriarch herself tends to move based on the **center of the clan**, maintaining the unity and balance of the group. This behavior ensures that the clan remains compact and moves as a single coordinated unit.

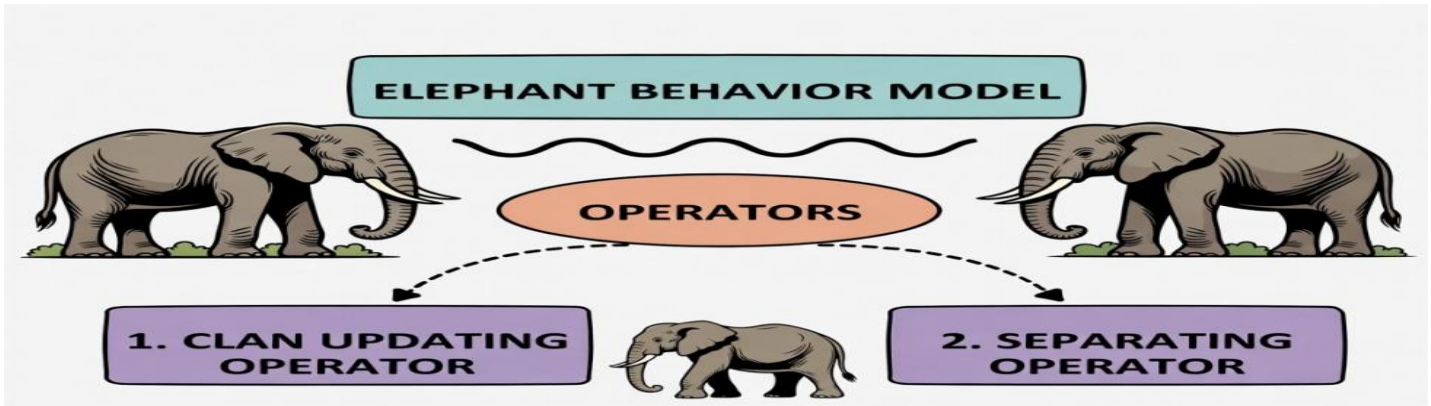


A crucial aspect of elephant social structure is the **separating behavior**. When male elephants reach adolescence (approximately 12–15 years old), they are forced to leave their natal clan. These young males either live alone or join small bachelor groups. This separation prevents inbreeding and encourages the exploration of new territories, thereby introducing diversity and expanding the search range of the overall population.

Modeling of Elephant Behavior in EHO:

The original EHO algorithm translates the above natural behaviors into two fundamental computational operators:





1. **Clan Updating Operator (Exploitation Phase)** This operator simulates the movement of elephants within each clan toward the matriarch and the influence of the clan center.
 - Non-matriarch elephants update their positions by moving toward the best elephant (matriarch) in their clan.
 - The matriarch updates her position based on the average position (center) of all elephants in the clan. This mechanism enhances local search capability and accelerates convergence toward promising solutions.
2. **Separating Operator (Exploration Phase)** This operator mimics the natural departure of adult male elephants from the clan. In each iteration, the worst-performing elephant in every clan is replaced by a completely random individual within the search space boundaries. This random replacement maintains population diversity, prevents premature convergence, and enables the algorithm to escape local optima.

By combining these two operators, EHO achieves an effective balance between **exploration** (global search) and **exploitation** (local search). The clan-based structure allows intensive searching around high-quality solutions, while the separating mechanism injects fresh randomness to explore distant regions of the search space.

This biologically inspired design makes EHO particularly suitable for complex, multimodal optimization problems commonly encountered in robotics path planning, trajectory optimization, and parameter tuning, where both local refinement and global diversity are essential.

3.2 Mathematical Equations:

To better understand how the Elephant Herding Optimization (EHO) algorithm works in practice, it is important to go through its mathematical formulation and the detailed steps of its implementation.

The following explanation focuses on how the algorithm operates step by step, including parameters and the main equations that govern the movement and updating of elephants within each clan

1. Solution Representation

Each elephant represents a candidate solution in a D-dimensional search space:

$$El_i = (x_{i1}, x_{i2}, \dots, x_{iD})$$

Where:

- (El_i) is the position of the (i^{th}) elephant
- (D) is the number of decision variables



2. Initialization

The initial population is generated randomly within the search space:

For each elephant i , each variable j

$$El_{i,j} = El_{min,j} + rand(0, 1) \times (El_{max,j} - El_{min,j})$$

Where:

- (El_{min}), (El_{max}) are lower and upper bounds
- ($rand(0,1)$) is a uniformly distributed random number

3. Fitness Evaluation

Each elephant is evaluated using a fitness function:

$$f(El_i)$$

The objective is to either minimize or maximize this function depending on the problem.

3.2.1 Clan Updating Operator (Exploitation)—Equation (1)

Within each clan, the position of elephants is influenced by the matriarch (best elephant).

Updating Normal Elephants

The new position of each elephant is computed as:

$$El_{new} = El_{old} + \alpha \times (El_{best} - El_{old}) \times R$$

Where:

- (El_{old}): current position
- (El_{new}): updated position
- (El_{best}): matriarch (best elephant in the clan)
- (α in $[0,1]$): scaling factor controlling influence
- (R in $[0,1]$): random number to increase diversity

☞ This equation ensures that elephants move toward the clan leader, enhancing exploitation.



3.2.2 Matriarch Updating Operator — Equation (2)

The matriarch is updated using the clan center:

$$El_{best}^{new} = \beta \times El_{center}$$

Where:

- (β in $[0,1]$): scaling factor
- (El_{center}): mean position of all elephants in the clan

Clan Center Calculation — Equation (3)

$$El_{center} = \frac{1}{n_{clan}} \sum_{j=1}^{n_{clan}} El_j$$



Where:

- (n_{clan}): number of elephants in the clan

☞ This helps stabilize the search and guide the leader toward the average behavior of the clan.

3.2.4 Separating Operator (Exploration) — Equation (4)

To maintain diversity and avoid local optima, the worst elephant in each clan is replaced.

$$El_{worst} = El_{min} + rand(0,1) \times (El_{max} - El_{min})$$

Where:

- (El_{worst}): worst-performing elephant
- (El_{min} , El_{max}): search space bounds

☞ This operator introduces new random solutions, improving exploration.



Pseudocode:

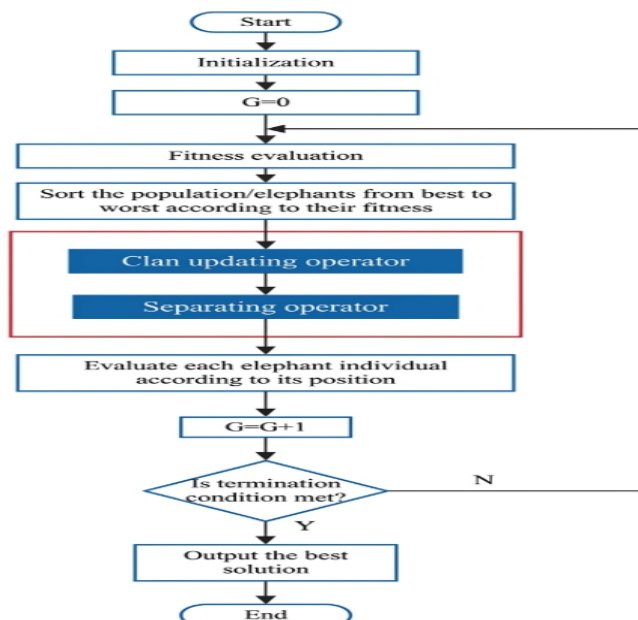
Algorithm: Elephant Herding Optimization (EHO)

```

1  Begin
2  Initialize population P randomly within [x_min, x_max]
3  Set max generation MaxGen,  $\alpha$ ,  $\beta$ , number of clans
4  Set generation counter t = 1
5  While t ≤ MaxGen do
6      Evaluate fitness of all elephants
7      Sort population by fitness (ascending for minimization)
8      Divide into clans
9      For each clan ci do
10         For each elephant j in clan ci do
11             If j is not matriarch (best in clan)
12                 Update xnew,ci,j using Equation (1)
13             Else (matriarch)
14                 Update xnew,ci using Equation (2) // via clan center Equation(3)
15             End if
16         Clip to bounds
17     End for
18 End for
19 For each clan ci do
20     Replace worst elephant using Equation (4)
21 End for
22 t = t + 1
23 End while
24 Return best solution
25 End

```

Flowchart:





Step-by-Step Algorithm Implementation:

The complete procedure of EHO can be described as follows:

Step 1: Initialization

- Initialize Algorithm parameters
- Generate (N) elephants randomly in search space

Step 2: Fitness Evaluation

- Compute ($f(El_i)$) for all elephants
- Sort the elephants from the best to the worst according to their fitness

Step 3: Divide elephants into clans

- Assign elephants to clans
- Identify the **matriarch** (best elephant) in each clan

Step 4: Clan Updating

For each clan:

1. Compute clan center: $El_center = \sum El_j / n_{clan}$
2. Update the matriarch: $El_best_new = \beta \times El_center$
3. Update each elephant: $El_new = El_old + \alpha \times (El_best - El_old) \times R$

Step 5: Separating Operator

- Identify the worst elephant in each clan
- Replace it using:

$$El_worst = El_min + rand(0,1) \times (El_max - El_min)$$

Step 6: Update Population

- Replace old positions with updated ones
- Recalculate fitness values

Step 7: Repeat

Repeat Steps 2–6 until:

- Maximum number of iterations is reached
or
- Convergence criterion is satisfied

Step 8: Termination

- Return the best solution found across all clans



Chapter (4): Numerical Example

4.1 Problem Definition

To demonstrate how the Elephant Herding Optimization (EHO) algorithm works in practice, we consider a simple optimization problem.

The objective is to minimize the following function:

$$f(x) = x^2$$

Where:

- x is a decision variable
- Search space: $-10 \leq x \leq 10$

The optimal solution is clearly at:

$$x = 0, f(x) = 0$$

4.2 Initialization:

Assume:

- Number of elements = 4
- Number of clans = 2
- Each clan has 2 elephants

Initial random positions:

Elephant	Position (X)	Fitness $f(x)=x^2$
E1	6	36
E2	-4	16
E3	3	9
E4	-7	49

4.3 Equations steps

Sorting helps identify the best elephants (matriarchs) in each clan.

After sorting (best $\rightarrow$ worst):

Elephant	Position (X)	Fitness $f(x)=x^2$	Rank
E3	3	9	1
E2	-4	16	2
E1	6	36	3
E4	-7	49	4

Divide into clans:

- **Clan 1** $\rightarrow$ (E3, E2)
- **Clan 2** $\rightarrow$ (E1, E4)

Matriarchs:

- **Clan 1** $\rightarrow$ E3 (best)



- **Clan 2** → E1

Clan Updating

Step 1: Calculate Clan Center

$$\text{Clan 1: } El_{center} = \frac{3+(-4)}{2} = -0.5$$

$$\text{Clan 2: } El_{center} = \frac{6+(-7)}{2} = -0.5$$

Step 2: Update Matriarch

Using:

$$El_{best,new} = \beta \times El_{center}$$

Assume:

$$\beta = 0.5$$

Clan 1:

$$E3_{new} = 0.5 \times (-0.5) = -0.25$$

Clan 2:

$$E1_{new} = 0.5 \times (-0.5) = -0.25$$

Step 3: Update Other Elephants

Using:

$$El_{new} = El_{old} + \alpha \times (El_{best} - El_{old}) \times R$$

Assume:

- $\alpha = 0.5$
- $R = 0.5$

Clan 1 (E2):

$$\begin{aligned} E2_{new} &= -4 + 0.5 \times (-0.25 - (-4)) \times 0.5 \\ &= -4 + 0.9375 \\ &= -3.0625 \end{aligned}$$

Clan 2 (E4):

$$\begin{aligned} E4_{new} &= -7 + 0.5 \times (-0.25 - (-7)) \times 0.5 \\ &= -7 + 1.6875 \\ &= -5.3125 \end{aligned}$$

Separating Operator

Replace worst elephant in each clan.

Worst elephants:

- Clan 1 → E2
- Clan 2 → E4

Using:

$$El_{worst} = El_{min} + rand(0,1) \times (El_{max} - El_{min})$$

Assume:

- $rand = 0.3$



$$El_{new} = -10 + 0.3 \times (20) = -10 + 6 = -4$$

Thus:

- $E2 = -4$
- $E4 = -4$

Updated Population

Elephant	New Position	Fitness
E3	-0.25	0.0625
E1	-0.25	0.0625
E2	-4	16
E4	-4	16

Result and Interpretation (After one iteration):

The solution has started to approach 0 (the optimal solution).

The best elephant (solution) is now:

$$x = -0.25$$

This shows that the algorithm is successfully converging toward the global optimum.

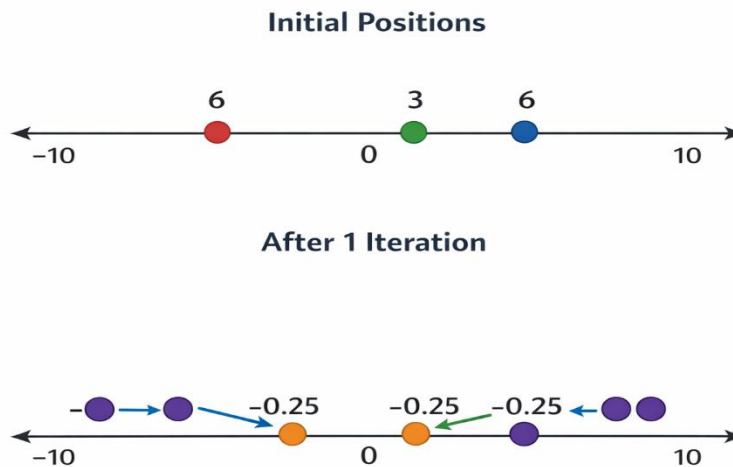


Figure: Movement of elephants before and after one iteration, showing convergence toward the optimal solution.

Conclusion of Example:

This numerical example demonstrates how the EHO algorithm:

- Improves solutions iteratively
- Moves elephants toward better positions (exploitation)
- Introduces randomness (exploration)



Chapter (5): Comparison & Conclusion

Metaheuristic optimization algorithms have been widely used to solve **complex optimization problems**. Among these algorithms, Elephant Herding Optimization (**EHO**) has gained attention due to its **simplicity** and **strong exploration capabilities**. To better understand its performance, it is useful to compare it with other well-known optimization techniques such as **Genetic Algorithm (GA)**, **Particle Swarm Optimization (PSO)**, and **Ant Colony Optimization (ACO)**. These algorithms share the same goal of finding optimal or near-optimal solutions but differ in their search strategies and performance characteristics.

One of the main evaluation metrics used to compare optimization algorithms is **convergence speed**. *Convergence speed* refers to how quickly an algorithm reaches a near-optimal solution. **EHO** demonstrates competitive *convergence speed* because the clan updating operator guides elephants toward **the best individual in each clan**. This mechanism accelerates exploitation and helps the population converge efficiently. Compared to Genetic Algorithm, which relies on **crossover** and **mutation operations**, **EHO** often requires fewer iterations to approach optimal solutions. Genetic Algorithm may spend more time exploring the search space due to its random mutation behavior.

Another important metric is **solution accuracy**. *Accuracy* measures how close the obtained solution is to the optimal value. **EHO** achieves good accuracy because it **balances** exploration and exploitation through clan updating and separating operators. Particle Swarm Optimization also provides high accuracy in many problems, but **PSO** can sometimes suffer from premature convergence, especially in **complex multimodal** search spaces. In contrast, **EHO** reduces this risk by replacing some individuals during the separating phase, which introduces diversity and prevents stagnation.

Exploration capability is another key factor when comparing optimization algorithms. *Exploration* refers to **the ability to search different regions of the solution space**. Algorithms with poor exploration may get trapped in local optima. Ant Colony Optimization performs strong exploration in discrete optimization problems but may not perform equally well in continuous domains. **EHO**, however, is flexible and can be applied to **both continuous and discrete problems**. The separating operator in **EHO** helps maintain diversity by randomly generating new elephants, which enhances global search capability.

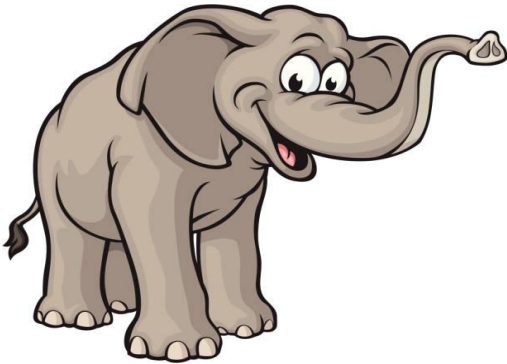
Computational complexity is also considered when evaluating optimization algorithms. Genetic Algorithm requires multiple operations such as selection, crossover, and mutation, which increases computational overhead. Similarly, Ant Colony Optimization requires pheromone updating and probability calculations. In comparison, **EHO** uses simpler updating rules, **which makes it easier to implement and computationally efficient**. This simplicity allows **EHO** to be applied to **large-scale problems with reduced computational cost**.

Another comparison metric is **robustness**. *Robustness* refers to the algorithm's ability to maintain performance across different types of problems. Particle Swarm Optimization is sensitive to parameter tuning, and improper parameter selection may affect performance. Genetic Algorithm also requires **careful tuning of crossover** and mutation rates. **EHO**, on the other hand, uses fewer control parameters, which simplifies **parameter tuning** and improves *robustness* across **different optimization problems**.

Despite its advantages, **EHO** has some limitations. Like other metaheuristic algorithms, its performance depends on parameter settings such as clan size and number of clans. In some cases, if diversity is not maintained, convergence may slow down. Researchers have proposed hybrid versions of **EHO** by combining it with other algorithms to further enhance performance. These hybrid approaches aim to improve exploration, convergence speed, and solution quality.



In conclusion, Elephant Herding Optimization is a **competitive metaheuristic algorithm** with several advantages over traditional optimization methods. It provides a **good balance** between exploration and exploitation, demonstrates **strong convergence speed**, and offers **high solution accuracy**. Compared to Genetic Algorithm, **Particle Swarm Optimization**, and **Ant Colony Optimization**, **EHO** is simpler to implement and requires fewer parameters. Additionally, its ability to **maintain diversity** helps prevent premature **convergence**. These characteristics make **EHO** suitable for solving complex optimization problems in engineering, machine learning, and operations research. Overall, the algorithm proves to be an effective and flexible optimization technique with promising performance across various evaluation metrics.



Chapter (6): References & citations

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